

Amitesh Puri

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West Bengal, India · 721433

Education

Manipal Academy of Higher Education (MIT Manipal)	Udupi, Karnataka, India
<i>M.Sc. in Applied Mathematics and Computing</i> CGPA: 7.28 / 10	Aug 2023 – Jun 2025
Ramnagar College, Vidyasagar University	Paschim Medinipur, West Bengal, India
<i>B.Sc. in Mathematics</i> Final Grade: 62%	Jul 2017 – Oct 2020
Kishorenagar Sachindra Siksha Sadan	Contai, West Bengal, India
<i>Higher Secondary (Class XII)</i> Final Grade: 74.5%	2014 – 2016

Research Experience

Addressing Conditional Shift in Graph Data: An Empirical Study of Distance Metrics [\[Report\]](#) [\[GitHub\]](#)
Master's Research Report, Manipal Academy of Higher Education Supervisor: Dr. Akansha Singh

- Studied the robustness of semi-supervised node classification under conditional distributional shift across several GNN architectures on benchmark datasets.
- Added distance metrics as auxiliary regularisers that penalise divergence between training and test node representations, isolating each metric's effect in controlled ablations.
- Used conformal prediction to evaluate coverage guarantees under distributional change.
- Implemented all experiments in PyTorch with fixed random seeds and publicly available code.

Research Projects

Physics-residual Flow Matching under distribution shift [\[Report\]](#) [\[GitHub\]](#)
Independent research report 2026

- Measured whether a physics-residual penalty improves a flow-matching generative model, and whether the conflict-free PBFM formulation is worth its added complexity over a plain weighted residual.
- Ran the comparison across five seeds, seven sampling budgets, and four graded distribution shifts, and reported the results without selection.
- Found that the weighted residual improved every in-distribution metric but did not hold up under shift, where it often reversed.

When does a geodesic probability path help latent flow matching? [\[Report\]](#) [\[GitHub\]](#)
Independent research report 2026

- Tested whether a straight-line or a geodesic probability path changes the outcome of latent flow matching, with the data, model, optimiser, and seed held fixed.
- Found the two paths equal across converged, seed-matched runs, a controlled null result.
- Showed through an ablation that path geometry helps only when the data lie on the assumed manifold.

Technical Training

Guided Projects Student Nov 2022 – Feb 2023
The Machine Learning Company Remote

- Completed applied projects in NLP and computer vision, including end-to-end pipelines for text classification, sentiment analysis, and language generation.
- Trained and evaluated deep learning models for image classification and object detection tasks.

Data Science Trainee

Evarcity

Aug 2021 – Jan 2022

Bangalore, India

- Built Python-based data analysis and machine learning workflows on real-world datasets, with practical exposure to feature engineering, model evaluation, and data visualisation.

Personal Projects

Cyberbullying Detection

[\[GitHub\]](#)

- Developed a text classifier in TensorFlow for identifying cyberbullying in social media content. Deployed the model through a Gradio interface, with user feedback collected and managed using Argilla for iterative data annotation.

Text Generation with GPT-2

[\[GitHub\]](#)

- Fine-tuned GPT-2 on a domain-specific corpus and deployed an interactive text generation interface for user evaluation and qualitative assessment of output quality.

English to Hindi Machine Translation

[\[GitHub\]](#)

- Fine-tuned a transformer-based sequence-to-sequence model for English to Hindi translation and deployed the system through a Streamlit application with integrated text-to-speech output.

Technical Skills

Programming	Python, R
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, PyCaret
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Data Visualization
Tools	Git, Gradio, Streamlit, Hugging Face Hub, Argilla